

# Supplementary Material for IAD-Risk

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## 1 Scope

This supplementary material documents the reproducibility boundary for the manuscript “IAD-Risk: Risk-Aware Identity-Agenda Disentanglement for Scholarly Work Deduplication.” It is intended to accompany the main manuscript, not to replace the project repository or the final artifact release. The repository contains executable code, fixture data, schema contracts, and build scripts; large third-party raw data, model checkpoints, and full experimental outputs are kept outside Git and must be distributed through an artifact release when numerical result reproduction is required.

## 2 Reproduction Levels

Table 1: Reproduction levels for auditing the repository and the reported evidence. L0/L1 levels verify executable code and fixture contracts, while L2/L3 levels are required for result-level numerical audit.

| Level                    | Required input                                                                                                                  | Audit target                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| L0 code check            | Source tree, package environment, and manuscript scripts                                                                        | Verify that the CLI, tests, public-release scan, and LaTeX build run in a clean environment.                               |
| L1 fixture rebuild       | Public miniature fixtures under <b>tests/fixtures</b>                                                                           | Verify that DeepMatcher, SciRepEval-style, OpenAlex-style, and IAD-Bench adapters produce the documented pair schema.      |
| L2 public-source rebuild | Independently obtained public raw files from the cited sources, plus <b>source_input_manifest</b> and <b>processing_run_log</b> | Rebuild derived IAD-Bench files with documented commands, input provenance, run metadata, output summaries, and checksums. |
| L3 result audit          | Released tables, predictions, logs, manifests, checksums, and commit identifiers                                                | Verify the numerical results reported in the manuscript against immutable artifact files.                                  |

## 3 Closest-Work Positioning

The main manuscript summarizes the novelty boundary in prose. Table 2 keeps the full closest-work positioning matrix auditable without interrupting the Related Work narrative.

Table 2: Positioning against the closest lines of work. IAD-Risk differs by treating agenda relatedness as a risk signal for merge decisions rather than as direct identity evidence.

| Line of work                         | Primary optimization target                                           | Limitation for scholarly deduplication                                            | IAD-Risk distinction                                                                        |
|--------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| End-to-end entity resolution systems | Blocking, matching, labeling, and deployment workflows.               | Workflow abstractions do not separate topical relatedness from work identity.     | Adds identity-agenda semantics and false-merge risk gates inside matching.                  |
| Neural entity matching               | Pair-level same-entity classification from tuple or text encoders.    | Strong text similarity can still link different papers in one agenda.             | Trains same-work, same-agenda, and agenda-non-identity heads with provenance-aware masking. |
| Scientific document representations  | Citation-informed or contrastive scientific relatedness embeddings.   | Relatedness embeddings support retrieval but are unsafe as direct merge evidence. | Uses representation similarity as candidate and agenda evidence, then blocks risky merges.  |
| Open scholarly metadata benchmarks   | Public metadata, citation, and topic signals for scalable evaluation. | Topic or citation proximity is silver evidence, not human identity gold.          | Reports gold, proxy, and silver strata separately and ties claims to label strength.        |

## 4 IAD-Bench Schema Contracts

The main manuscript summarizes the schema in prose. The full tables below define the required document and pair fields used by fixture rebuilding, public-source rebuilding, and artifact validation.

Table 3: IAD-Bench document schema contract. Missing values are represented by empty strings, empty arrays, or null values rather than by dropping required fields.

| Field group            | Required fields                                                                                                   | Interpretation role                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Record identity        | <code>document_id</code> , <code>source_dataset</code>                                                            | Makes each standardized record traceable to its originating public source.                  |
| Text and authorship    | <code>title</code> , <code>abstract</code> , <code>authors</code>                                                 | Provides the textual and author evidence used by identity and agenda features.              |
| Bibliographic metadata | <code>year</code> , <code>venue</code> , <code>doi</code> , <code>arxiv_id</code> , <code>openalex_work_id</code> | Supplies identifier and publication-context evidence for same-work decisions.               |
| Agenda context         | <code>topics</code> , <code>references</code>                                                                     | Supplies topical and citation evidence used for same-agenda and hard-negative construction. |

Table 4: IAD-Bench pair schema contract. The schema records identity labels, agenda labels, provenance, split assignment, and hard-negative status for each pair.

| Field group        | Required fields                                                                                   | Interpretation role                                                                                      |
|--------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Pair identity      | <code>pair_id</code> , <code>source_document_id</code> ,<br><code>target_document_id</code>       | Makes every row traceable to two standardized documents and prevents anonymous aggregate-only reporting. |
| Relation targets   | <code>relation_label</code> , <code>expected_label</code> ,<br><code>expected_agenda_label</code> | Separates same-work identity from agenda relatedness before metrics are computed.                        |
| Evidence source    | <code>label_source</code> , <code>label_strength</code> ,<br><code>label_provenance</code>        | Records whether a label is gold, proxy, silver, or human-audit evidence and what signals support it.     |
| Evaluation control | <code>split</code> , <code>hard_negative_level</code>                                             | Controls train/dev/test/audit use and identifies agenda-level hard negatives for HNFMR.                  |

## 5 Method Design Boundaries

The main manuscript summarizes method-design boundaries in prose. The tables below keep the failure-control rationale, risk-score rationale, design alternatives, feature-head specification, operational net-benefit, and version-identifier policies auditable without overloading the Method narrative.

Table 5: Failure-control rationale of IAD-Risk. Each component targets a concrete way in which agenda-level similarity can be mistaken for work identity.

| Failure pathway                                               | Design response                                                                         | Remaining boundary                                                     |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Topically close papers receive high semantic similarity.      | Separate agenda evidence from identity evidence before the merge gate.                  | Requires the benchmark to include agenda-level hard negatives.         |
| Silver metadata is treated as if it were human gold.          | Preserve label strength and provenance, and mask supervision by valid label source.     | Silver labels support stress testing, not exhaustive truth claims.     |
| Pairwise errors contaminate clusters through transitivity.    | Apply risk gating before constrained union-find and respect cannot-link evidence.       | Cluster-level guarantees require complete cannot-link coverage.        |
| Thresholds turn a classifier into an unsafe automatic merger. | Expose work, ANI, and risk thresholds and evaluate held-out merge decisions.            | Threshold transfer should be rechecked under new source distributions. |
| Proxy labels are over-interpreted.                            | Keep proxy rows separate unless the label source supports the relation being evaluated. | Proxy rows remain non-human evidence even when reproducible.           |

Table 6: Risk score design rationale. The score is designed as a conservative merge gate, not as an unconditional probability of error.

| Design element                                | Design rationale                                                                | Boundary                                                                           |
|-----------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| $p_{\text{ani}}$ term                         | Directly represents evidence that a related pair is not the same work.          | Requires valid ANI supervision or artifact-backed hard-negative evidence.          |
| $p_{\text{agenda}}(1 - p_{\text{work}})$ term | Flags pairs that are topically related but lack corresponding identity support. | It is not a calibrated probability unless validated against held-out artifacts.    |
| Max operator                                  | Makes either risk pathway sufficient to block a merge.                          | Conservative blocking can increase deferrals and must be tuned to the application. |
| Threshold gate                                | Converts the risk proxy into a merge, block, or defer decision.                 | Threshold transfer must be rechecked under new source distributions.               |

Table 7: Design alternatives considered during method design. Each rejected shortcut lacks an explicit mechanism for separating agenda relatedness from work identity or for auditing false-merge risk.

| Alternative                                         | Why it is insufficient for this failure mode                                                                                     | IAD-Risk design response                                                                         | Evidence boundary                                                     |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Tune only a representation-similarity threshold.    | High scientific relatedness can still connect different works in the same task, benchmark, or citation community.                | Use representation similarity as candidate and agenda evidence, not direct merge evidence.       | Supports false-merge-risk motivation, not a full retrieval benchmark. |
| Rely on one supervised pair classifier.             | A strong classifier can reduce false merges but does not expose whether a positive score comes from identity or agenda evidence. | Predict same-work, same-agenda, and agenda-non-identity signals separately before gating.        | RoBERTa remains a strong baseline; broad superiority is not claimed.  |
| Use provenance as a model feature.                  | Source or split identifiers can create leakage and make held-out interpretation fragile.                                         | Keep provenance as audit metadata and apply provenance-aware supervision masking.                | Source-heldout claims require stronger source coverage.               |
| Force every candidate into a binary merge decision. | Some pairs have related evidence but insufficient identity evidence, making automatic merging risky.                             | Emit merge, block, or defer through the risk-aware gate.                                         | Defer behavior needs deployment-specific review policies.             |
| Select thresholds after test results.               | Post-hoc threshold choice can hide false-merge risk and inflate held-out conclusions.                                            | Fix operating points before final reporting and require threshold artifacts for stronger claims. | Threshold stability needs a released grid and checksums.              |

Table 8: Feature and head specification for IAD-Risk transformer variants. Metadata fields are retained for auditing but are not used as model features.

| Head or stage  | Main input fields                                                                                                                                           | Supervision or calculation                                                                    | Output role                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Identity head  | Transformer distances, title similarity, author overlap, first-author match, venue similarity, year match, DOI/arXiv/OpenAlex identifier agreement.         | Same-work labels from valid identity evidence, with provenance-aware masking.                 | Estimates $p_{\text{work}}$ as positive merge evidence.                           |
| Agenda head    | Transformer cosine, title and venue similarity, OpenAlex-style topic overlap, reference Jaccard similarity, year-distance score.                            | Same-agenda proxy or silver evidence when the source supports agenda relatedness.             | Estimates $p_{\text{agenda}}$ as relatedness evidence, not direct merge evidence. |
| ANI risk head  | Transformer distances, title and author signals, topic overlap, reference overlap, year conflict, identifier agreement, and different-identifier conflicts. | Agenda-non-identity hard negatives from valid hard-negative sources.                          | Estimates $p_{\text{ani}}$ as direct blocking evidence.                           |
| Risk gate      | $p_{\text{work}}$ , $p_{\text{agenda}}$ , $p_{\text{ani}}$ , $\tau_w$ , $\tau_n$ , $\tau_r$ , and cannot-link flags.                                        | Computes $p_{\text{risk}} = \max\{p_{\text{ani}}, p_{\text{agenda}}(1 - p_{\text{work}})\}$ . | Emits merge, block, or defer decisions.                                           |
| Audit metadata | Pair ID, source pair ID, label strength, label source, provenance, split, and hard-negative level.                                                          | Copied from pair records and result artifacts.                                                | Supports artifact review; it is not a training feature.                           |

Table 9: Training and inference trace for IAD-Risk. Each phase states the required operation and the artifact or invariant that makes the phase auditable.

| Phase              | Required operation                                                                                       | Auditable output or invariant                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Schema loading     | Load pair records with relation labels, label strength, provenance, split, and hard-negative fields.     | Pair IDs and split fields remain unchanged from input to prediction output.                      |
| Supervised fitting | Optimize the masked objective only where the relevant supervision mask is valid.                         | Gold, proxy, and silver labels are not silently converted into labels for unrelated heads.       |
| Threshold fixation | Select or record $\tau_w$ , $\tau_n$ , and $\tau_r$ before held-out reporting.                           | Threshold source, value, selection split, and selection metric are written to the threshold log. |
| Pair scoring       | Predict $p_{\text{work}}$ , $p_{\text{agenda}}$ , and $p_{\text{ani}}$ , then derive $p_{\text{risk}}$ . | Prediction rows expose relation scores and the derived risk value for each pair.                 |
| Decision emission  | Apply cannot-link evidence and the risk-aware merge gate to emit merge, block, or defer.                 | Decision rows record threshold values, threshold source, and cannot-link status.                 |
| Metric export      | Count same-work F1, FMR, HNFMR, coverage, block rate, and defer rate from the same decision file.        | Metric summaries and checksums bind denominators to the prediction artifact.                     |

Table 10: Scoring and merge algorithm for IAD-Risk. The table gives the execution order that connects pair inputs, relation-head outputs, risk gating, and metric-counting artifacts.

| Step | Operation                                                        | Required inputs or outputs                                                           | Audit role                                                          |
|------|------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1    | Load a candidate pair and schema metadata.                       | Pair ID, document IDs, label strength, provenance, split, and hard-negative level.   | Fixes row identity before scoring.                                  |
| 2    | Build feature groups without using audit metadata as predictors. | Identity, agenda, ANI, and cannot-link feature groups.                               | Prevents provenance or split leakage.                               |
| 3    | Predict relation heads.                                          | $p_{\text{work}}$ , $p_{\text{agenda}}$ , and $p_{\text{ani}}$ .                     | Records interpretable relation evidence.                            |
| 4    | Compute derived false-merge risk.                                | $p_{\text{risk}} = \max\{p_{\text{ani}}, p_{\text{agenda}}(1 - p_{\text{work}})\}$ . | Makes the blocking score reproducible.                              |
| 5    | Apply the merge gate and cannot-link evidence.                   | $\tau_w$ , $\tau_n$ , $\tau_r$ , cannot-link flag, and threshold source.             | Emits merge, block, or defer under a fixed operating point.         |
| 6    | Write prediction and metric audit fields.                        | Decision, row scope, denominators, thresholds, and checksum-bound artifact row.      | Supports same-work F1, FMR, HNFMR, coverage, and defer-rate audits. |

Table 11: Operational complexity and net-benefit boundary. The framework is intended for safety-sensitive scholarly identity decisions, not as a universal replacement for simple deduplication pipelines.

| Operational item     | Required practice                                                                                                           | Net-benefit boundary                                                                                                                        |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Model complexity     | Train and audit same-work, same-agenda, and ANI heads under one pair schema.                                                | The net benefit is strongest in high-stakes scholarly indexes where false merges are costly.                                                |
| Threshold governance | Select shared thresholds on validation evidence and record them in artifacts.                                               | Thresholds are not tuned per pair or changed after test inspection.                                                                         |
| Deferral policy      | Record the deferral budget and manual-review capacity before deployment.                                                    | A high defer rate is acceptable only when review capacity exists.                                                                           |
| Automation claim     | Compare automatic merge coverage, block rate, defer rate, and manual-review capacity under the predeclared operating point. | Without workload evidence, the method should be interpreted as a conservative safety filter rather than as a throughput-improving pipeline. |
| Simple alternatives  | Retain simple rules for low-risk bulk cleanup or identifier-only reconciliation.                                            | IAD-Risk is not a universal replacement for simple deduplication pipelines.                                                                 |

Table 12: Version and identifier boundary for merge decisions. The policy prevents related scholarly versions or conflicting identifiers from being silently collapsed into one work identity.

| Evidence condition   | Operational policy                                                                                        | Boundary statement                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Identifier agreement | DOI, arXiv, or OpenAlex agreement is positive identity evidence when other metadata do not contradict it. | identifier agreement supports merge eligibility, but it does not replace source and metadata checks. |
| Identifier conflict  | Conflicting DOI, arXiv, or OpenAlex fields are treated as risk evidence before any automatic merge.       | identifier conflict creates cannot-link or defer evidence unless an audit log resolves the conflict. |
| Version relationship | Related preprint, conference, and journal versions require an explicit version policy.                    | not every related version is automatically the same work.                                            |
| Ambiguous lineage    | When lineage cannot be established from identifiers and metadata, the pair is not forced into a merge.    | manual adjudication is required for ambiguous version boundaries.                                    |
| Cluster propagation  | Version decisions are recorded before transitive clustering.                                              | version policy must be declared before cluster-level merging.                                        |

## 6 Environment Setup

The repository is designed to run from a standard Python environment without committing raw third-party data. Reviewers can validate the code and fixture-level reproduction path from the project root with the following commands:

```
conda create -n iad-sieve python=3.11 -y
conda activate iad-sieve
python -m pip install -e .
python -m iad_sieve.cli --help
python scripts/check_public_release.py
python manuscript/scripts/verify_fixture_rebuild.py
```

The environment check does not download full raw datasets. It verifies package installation, CLI availability, public-release boundaries, and no-network fixture rebuilding. Full numerical result reproduction still requires the L2/L3 data and artifact requirements described below.

## 7 No-Network Fixture Rebuild

The fixture rebuild validates the data adapters and IAD-Bench schema without depending on external network access. It is not a substitute for the full paper experiment.

```
python -m iad_sieve.cli prepare-deepmatcher \
  --table-a tests/fixtures/deepmatcher/tableA.csv \
  --table-b tests/fixtures/deepmatcher/tableB.csv \
  --pairs tests/fixtures/deepmatcher/test.csv \
  --dataset-name deepmatcher_fixture \
  --output-dir outputs/repro_fixture/deepmatcher

python -m iad_sieve.cli prepare-scirepeval-proximity \
  --metadata tests/fixtures/scirepeval/metadata.jsonl \
  --pairs tests/fixtures/scirepeval/scidocs_cite_pairs.csv \
  --dataset-name scirepeval_fixture \
```

```

--output-dir outputs/repro_fixture/scirepeval

python -m iad_sieve.cli prepare-openalex-weak-labels \
  --works tests/fixtures/openalex/works.jsonl \
  --citations tests/fixtures/openalex/coci.csv \
  --dataset-name openalex_fixture \
  --output-dir outputs/repro_fixture/openalex \
  --min-shared-references 1 \
  --max-pairs 20

python -m iad_sieve.cli build-iad-bench \
  --source-dirs \
    outputs/repro_fixture/deepmatcher \
    outputs/repro_fixture/scirepeval \
    outputs/repro_fixture/openalex \
  --output-dir outputs/repro_fixture/iad_bench \
  --seed 42

```

The manuscript package includes a verification wrapper for the same no-network path:

```
python manuscript/scripts/verify_fixture_rebuild.py
```

The wrapper runs the four fixture commands in a temporary directory and checks that each source adapter writes `eval_documents.jsonl`, `eval_pairs.jsonl`, and `dataset_summary.jsonl`, and that the assembled IAD-Bench package writes pair, document, split, summary, provenance, and dataset-card files.

## 8 Public-Source Rebuild

For full data reconstruction, raw public files should be obtained from their original sources and placed under local `data/raw/` directories. These directories are intentionally excluded from Git. The project keeps conversion commands and schema contracts under version control, while source licenses and access conditions remain governed by the original providers.

```

python -m iad_sieve.cli prepare-deepmatcher \
  --table-a data/raw/deepmatcher/DBLP-ACM/tableA.csv \
  --table-b data/raw/deepmatcher/DBLP-ACM/tableB.csv \
  --pairs data/raw/deepmatcher/DBLP-ACM/test.csv \
  --dataset-name DBLP-ACM \
  --output-dir data/processed/eval_sources/deepmatcher_dblp_acm

python -m iad_sieve.cli prepare-scirepeval-proximity \
  --metadata data/raw/scirepeval/metadata.jsonl \
  --pairs data/raw/scirepeval/scidocs_cite_pairs.csv \
  --dataset-name scidocs_cite \
  --output-dir data/processed/eval_sources/scirepeval_scidocs_cite

python -m iad_sieve.cli prepare-openalex-weak-labels \
  --works data/raw/openalex/works_cs_sample.jsonl \
  --citations data/raw/opencitations/coci.csv \
  --dataset-name openalex_cs_sample \
  --output-dir data/processed/eval_sources/openalex_cs_sample \

```

```

--min-shared-references 1 \
--max-pairs 5000

python -m iad_sieve.cli build-iad-bench \
--source-dirs \
  data/processed/eval_sources/deepmatcher_dblp_acm \
  data/processed/eval_sources/scirepeval_scidocs_cite \
  data/processed/eval_sources/openalex_cs_sample \
--output-dir data/processed/iad_bench/open_v3 \
--train-ratio 0.8 \
--dev-ratio 0.1 \
--seed 42

```

## 9 Public-Source Rebuild Audit Boundary

The L2 public-source rebuild should be treated as auditable only when the raw-input boundary, processing run, and derived-output boundary are recorded together. A reviewer does not need the raw third-party files in Git, but should be able to inspect a release-level `source_input_manifest`, a `processing_run_log`, and output summaries that identify which public files, commands, seeds, software versions, and checksums produced each derived evaluation source.

Table 13: Audit boundary for L2 public-source rebuilds. The table states the files that should accompany a rebuild when raw third-party data are not committed to Git.

| Audit item                         | Required content                                                                                                                                                            | Audit use                                                                                                                                 |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <code>source_input_manifest</code> | Source name, acquisition date or version, original provider, local file name, record count when known, license boundary, and SHA256 checksum for each input file.           | Shows that the rebuild used identifiable public inputs without redistributing those inputs in Git.                                        |
| <code>processing_run_log</code>    | Command line, code commit, environment summary, random seed, start and finish timestamps, input manifest reference, output path, and exit status for each processing stage. | Shows that DeepMatcher, SciRepEval-style, OpenAlex/OpenCitations, and IAD-Bench assembly commands were run under a fixed source revision. |
| Output summaries                   | Dataset summaries, split summaries, provenance summaries, row counts, label-strength counts, and checksum entries for derived files.                                        | Lets reviewers compare derived evaluation artifacts against the manuscript scope before using them as numerical evidence.                 |
| Exclusion record                   | Explicit statement that raw third-party files, credentials, caches, model checkpoints, and local experiment outputs are excluded from the public Git package.               | Prevents a clean repository from being mistaken for an incomplete or hidden data release.                                                 |

The artifact release template therefore requires `source_input_manifest` and `processing_run_log` in addition to result tables, predictions, threshold logs, and split summaries. These files do not upgrade fixture-level reproduction into a full numerical audit by themselves. They provide the chain of custody needed to decide whether an L2 rebuild is consistent with the L3 result artifacts used to support manuscript numbers.

## 10 Data Processing Traceability

Table 14 gives the audit trace from public source files to derived evaluation artifacts. The repository commits the processing commands and schema checks, but keeps the raw source files and large derived outputs outside Git. This separation allows an independent reader to rebuild the same artifact classes from obtained public sources while respecting source licenses and keeping the public repository small.

Table 14: Traceability from public data sources to derived IAD-Bench artifacts. Git contains commands, schemas, tests, and manifests; full-size inputs and outputs are verified through external artifact checksums.

| Stage                | Local input boundary                                 | Processing entry point                        | Expected output                                                 | Git boundary                             |
|----------------------|------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------|------------------------------------------|
| DeepMatcher gold     | Raw benchmark tables and pair labels.                | <code>prepare-deepmatcher.</code>             | Eval documents, eval pairs, and dataset summary.                | Raw tables excluded; fixtures tested.    |
| SciRepEval proxy     | Public metadata and citation-pair files.             | <code>prepare-sciреeval-proximity.</code>     | Proxy agenda pairs and source summaries.                        | Full metadata excluded; schema retained. |
| OpenAlex weak labels | OpenAlex works and OpenCitations COCI files.         | <code>prepare-openalex-weak-labels.</code>    | Silver hard negatives with provenance.                          | Large sources excluded; rules retained.  |
| IAD-Bench assembly   | Processed source directories from previous stages.   | <code>build-iad-bench.</code>                 | Unified documents, pairs, splits, provenance, and dataset card. | Full package released as L2/L3 artifact. |
| Result audit         | Prediction, score, threshold, log, and metric files. | Artifact finalization and validation scripts. | Manifest, checksums, tables, predictions, and command logs.     | External artifact required.              |

## 11 Artifact Package Requirements

Full numerical result reproduction should use a released artifact package rather than raw Git-tracked files. A complete artifact package should contain a manifest, checksums, tables, predictions, configuration files, logs, and the commit identifier used to generate the results.

```
paper-artifacts/
  README.md
  manifest.json
  checksums.sha256
  configs/
  tables/
  predictions/
  reports/
  logs/
```

The manifest should record the source data version, processing commands, dependency versions, random seeds, thresholds, split configuration, hardware summary, and SHA256 checksum for every

result file. The checksums should be validated before using artifact files to support claims in the manuscript.

Table 15: Result artifact crosswalk for the Open-v2 evidence snapshot. Each row family in the main table has a required artifact path or manifest entry before it can be used for full numerical audit.

| Main-table evidence                 | Required artifact IDs                                                                                    | Audit purpose                                                                                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Open-v2 aggregate table             | <code>open_v2_main_results</code> ,<br><code>iad_bench_split_summary</code>                              | Verifies row scopes, pair counts, document counts, label strata, metric columns, and selective-decision coverage fields.                                             |
| L2 public-source rebuild provenance | <code>source_input_manifest</code> ,<br><code>processing_run_log</code>                                  | Verifies public input provenance, processing commands, code commit, environment summary, random seed, output path, exit status, and checksum-fixed rebuild metadata. |
| Representation cosine baselines     | <code>representation_baseline_scores</code> ,<br><code>threshold_selection_logs</code>                   | Verifies SciNCL and SPECTER2 scores, threshold choices, and full-pair false-merge rates.                                                                             |
| RoBERTa pair classifier             | <code>supervised_baseline_predictions</code> ,<br><code>threshold_selection_logs</code>                  | Verifies supervised baseline predictions under the same pair schema and fixed operating point.                                                                       |
| IAD-Risk transformer variants       | <code>iad_risk_predictions</code> ,<br><code>threshold_selection_logs</code>                             | Verifies held-out predictions, relation-head scores, risk-gate decisions, thresholds, and split identifiers.                                                         |
| Optional stronger claims            | <code>bootstrap_intervals</code> , <code>ablation_suite</code> ,<br><code>manual_validation_slice</code> | Required only before claiming confidence intervals, component causality, or human label precision.                                                                   |
| Threshold-stability claims          | <code>threshold_sensitivity_grid</code> ,<br><code>threshold_selection_logs</code>                       | Required only before claiming stability across operating points or threshold ranges.                                                                                 |

## 12 Selective Decision Coverage Boundary

The main manuscript summarizes the selective-decision coverage boundary in prose. Table 16 preserves the coverage and review-load quantities that must be released before false-merge control can be read as operational automation evidence. Let  $N$  denote the evaluated pair count for one row of the result table, with  $M$  automatic merges,  $B$  blocked pairs, and  $D$  deferred pairs under the same fixed operating point. The release artifact should satisfy  $N = M + B + D$  for every row. Automatic merge coverage is  $AMC = M/N$ , block rate is  $BR = B/N$ , and defer rate is  $DR = D/N$ . Let  $B_{\text{review}} \leq B$  and  $D_{\text{review}} \leq D$  denote the blocked and deferred pairs that the declared workflow sends to human review; terminal cannot-link blocks that need no further adjudication remain counted in  $B$  but are not counted in  $R$ . If  $R = B_{\text{review}} + D_{\text{review}}$  and  $C$  is the predeclared review capacity for the batch, capacity-normalized review load is  $CNRL = R/C$ . These quantities must be derived from the same prediction artifact, threshold configuration, row scope, and denominator record as FMR and HNFMR.

Table 16: Selective decision coverage boundary. Coverage and review-load quantities must be computed from the same prediction files as the reported risk metrics before operational deployment claims are made.

| Quantity                        | Required artifact source                                                                                         | Interpretation boundary                                                                                                                                  |
|---------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Automatic merge coverage        | Merge decisions from the same prediction files and thresholds.                                                   | Needed before claiming that the system resolves a large fraction of pairs automatically.                                                                 |
| Block rate                      | Blocked decisions under the reported risk gate.                                                                  | Separates safety-driven blocking from improved identity recognition.                                                                                     |
| Defer rate                      | Deferred decisions under the fixed operating point.                                                              | High defer rate shifts work to manual review and must be reported with FMR/HNFMFMR.                                                                      |
| Review load                     | Counts of review-required blocked and deferred pairs by label stratum, source, and hard-negative level.          | Terminal safety blocks must be separated from pairs that require human handling before claiming operational throughput reduction or review-cost savings. |
| Capacity-normalized review load | Review workload after subtracting automatic merges and counting only blocked or deferred pairs requiring review. | Required before interpreting low false-merge rates as operational cost savings.                                                                          |

## 13 Pair-to-Cluster Evidence Boundary

The main manuscript summarizes the pair-to-cluster evidence boundary in prose. Table 17 preserves the artifact requirements that separate pair-level risk control from cluster-level deployment claims.

Table 17: Pair-to-cluster evidence boundary. Cluster-level deployment claims require artifacts beyond pair-level prediction and metric files.

| Evidence item            | Required artifact source                                                                                                  | Interpretation boundary                                                             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| cluster assignments      | Released component or cluster ID for each evaluated record.                                                               | Needed before computing cluster purity, split errors, or merged-work contamination. |
| cannot-link violations   | <code>cannot_link_audit</code> generated from the same merge decisions and identifier-conflict rules.                     | Needed before claiming that risk gates prevent transitive false merges.             |
| Cluster metric summary   | <code>cluster_metric_summary</code> with cluster contamination rate, singleton rate, merge coverage, and conflict counts. | Needed before claiming cluster-level quality rather than pair-level risk control.   |
| Pair-to-cluster trace    | Mapping from pair decisions, thresholds, and cannot-link flags to final components.                                       | Required to audit whether a pair-level error changes a downstream cluster.          |
| Current manuscript claim | Pair-level F1, FMR, and HNFMFMR under fixed operating points.                                                             | The paper does not claim cluster-level contamination is eliminated.                 |

```
python manuscript/scripts/build_artifact_release_skeleton.py \
  --output-dir /path/to/release \
  --repository-commit <commit>
python manuscript/scripts/populate_artifact_release.py \
  --artifact-dir /path/to/release \
  --source-dir /path/to/source-artifacts \
```

```

--preflight-only
python manuscript/scripts/populate_artifact_release.py \
--artifact-dir /path/to/release \
--source-dir /path/to/source-artifacts
python manuscript/scripts/finalize_artifact_release.py \
--artifact-dir /path/to/release
python manuscript/scripts/validate_artifact_release.py \
--artifact-dir /path/to/release

```

The release preparation commands create a scaffold, check source artifact completeness before copying, copy generated result files from a source artifact directory, refresh manifest and checksum metadata, and then validate the final package. The `--preflight-only` command reuses the release manifest copy plan to check required source artifact paths and optional mapping paths without copying files, writing `artifact_population_log.jsonl`, or running finalization. A passing source preflight shows that the source artifact directory is structurally ready for population; it does not prove row-level schemas, checksums, public accessibility, source-control commit binding, or numerical correctness. The validation command checks the manifest schema, required artifact files, required result identifiers, checksum coverage, repository commit metadata, claim-boundary flags, conditional claim artifacts, and exclusion of raw third-party data, caches, credentials, local outputs, and model checkpoints.

The main Open-v2 result artifact is `open_v2_main_results`. Its CSV file must include per-row denominator counts, the per-row threshold source, the scope label used in the main table, automatic merge count, block count, defer count, automatic merge coverage, defer rate, and capacity-normalized review load. This row-level schema lets reviewers trace each reported same-work F1, FMR, and HNFMR value to the evaluated pair scope, denominator, operating-point source, selective-decision coverage, workload boundary, and fixed artifact version.

The prediction and threshold artifacts must be row-auditable rather than opaque model dumps. The `iad_risk_predictions` JSONL file must include system, `pair_id`, `source_document_id`, `target_document_id`, expected labels, label strength, hard-negative level, split identifiers, relation-head scores, `work_threshold`, `agenda_block_threshold`, `risk_threshold`, threshold source, and `merge_prediction`. The `representation_baseline_scores` JSONL file must include the same pair and label fields plus normalized score, `score_field`, `threshold_value`, threshold source, and `merge_prediction`. The `supervised_baseline_predictions` JSONL file must include the same pair and label fields plus `match_probability`, `threshold_value`, threshold source, and `merge_prediction`. The `threshold_selection_logs` JSONL file must include system, `threshold_name`, `threshold_value`, `selection_split`, `selection_metric`, `selection_rule`, `applied_scope`, and `score_field` so that fixed operating points cannot be selected from the test rows after evaluation.

The artifact manifest uses conditional claim flags so that optional stronger claims cannot be enabled without their evidence files. Confidence-interval claims require `bootstrap_intervals`; component-causality claims require `ablation_suite`; human-validation claims require `manual_validation_slice`; threshold-stability claims require `threshold_sensitivity_grid`; cluster-level quality claims require `cluster_metric_summary` and `cannot_link_audit`; and broad method-ranking claims require the interval, validation, and threshold-grid artifacts together. If a conditional flag is set to true, the corresponding artifact row must be marked required and must include a file path and SHA256 checksum.

Cluster-level quality claims require evidence beyond pair-level prediction files. The `cluster_metric_summary` artifact should include cluster assignments, cluster contamination rate, singleton rate, merge coverage, conflict counts, and a pair-to-cluster trace from accepted links to connected components. The `cannot_link_audit` artifact should report cannot-link coverage, violations, un-

resolved conflicts, and identifier-conflict rules under the same threshold and merge policy used for the reported results. Without both files and their checksums, the manuscript should discuss pair-level false-merge control only and should not claim cluster-level contamination reduction.

## 14 Split and Leakage Controls

The main manuscript summarizes split and leakage controls in prose. Table 18 preserves the controls used to interpret IAD-Bench results and the stronger evidence required before broader held-out or generalization claims are made.

Table 18: Split and leakage controls used to interpret IAD-Bench results. The controls define what the current evidence supports and what stronger follow-up evidence would require.

| Control                               | Manuscript role                                                                                                                                                                           | Stronger evidence boundary                                                                                                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Train/dev/test split field            | Separates fitting, threshold selection, and held-out reporting.                                                                                                                           | Full numerical reproduction requires released prediction files, configs, and split summaries.                                                                                                                                                |
| Unordered pair leakage guard          | Detects whether the same document pair appears across multiple splits.                                                                                                                    | A blocked leakage audit prevents held-out claims until duplicate pair assignments are removed.                                                                                                                                               |
| Document/cluster split-overread guard | Treats the current split as pair-record held-out evidence unless source artifacts prove document-disjoint or cluster-disjoint grouping before split assignment.                           | Document-disjoint, cluster-disjoint, or unseen-source generalization requires grouped split manifests, document/cluster overlap reports, per-scope denominators, prediction checksums, threshold logs, and command provenance.               |
| Label-stratum coverage audit          | Checks whether gold, proxy, and silver evidence appear in the intended evaluation strata.                                                                                                 | Missing strata must be reported as evidence gaps rather than averaged into a single score.                                                                                                                                                   |
| Source-heldout readiness audit        | Tests whether relation labels have enough independent label sources for source-heldout claims and whether train, validation, and held-out source partitions were declared before fitting. | Broad source generalization remains conditional until released artifacts exclude source identifiers from predictive features and report per-source denominators, command logs, split summaries, prediction checksums, and threshold records. |
| Topic-heldout readiness audit         | Tests whether enough topics exist to reserve unseen-topic evaluation slices.                                                                                                              | Cross-topic stability should not be claimed when topic coverage is insufficient.                                                                                                                                                             |

## 15 Baseline Audit Boundary

The main manuscript reports the baseline decision rule in prose to keep the result narrative readable. The full audit matrices below define which baseline families enter the primary result table and which comparison controls must hold before a broad ranking claim is made.

Table 19: Baseline inclusion rationale. A baseline enters the primary result table only when it has auditable outputs under the reported pair scope.

| Baseline family                       | Current role                                                                    | Inclusion or exclusion rule                                                                                                                           |
|---------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exact identifier matching             | Important sanity-check family for DOI, arXiv, and publication-lineage evidence. | Excluded from primary result table when only utility code or fixture-level checks are available; full metrics require prediction files and checksums. |
| Title-normalization rules             | Useful lower-bound family for metadata-only deduplication.                      | Excluded until the same pair scope, threshold records, metric summaries, and error cases are released.                                                |
| Traditional entity-resolution systems | Relevant workflow baselines for blocking, matching, and clustering comparisons. | Included as primary evidence only when metric summaries, prediction files, threshold records, and checksums are available.                            |
| Scientific representation baselines   | Primary stress-test evidence for single-space scientific relatedness.           | Included because SciNCL cosine and SPECTER2 adapter cosine have reported score summaries under Open-v2.                                               |
| RoBERTa pair classification           | Strong supervised comparator for pair-level matching.                           | Included as a primary supervised comparator while keeping the result below broad ranking claims.                                                      |

Table 20: Baseline fairness controls for the Open-v2 evidence snapshot. The controls make the reported comparison auditable while preserving the scope boundary of each row family.

| Control               | Application in the reported table                                                                                                | Interpretation boundary                                                               |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Input rows            | Baselines use the same IAD-Bench pair records within each reported row scope.                                                    | A stricter ranking requires same-scope released prediction files for all rows.        |
| Split use             | Trainable rows use the same train/dev/test split field; non-trainable representation rows are reported as fixed score baselines. | Different available scopes should not be read as a single comparative ranking.        |
| Threshold selection   | Threshold-sensitive rows use validation-selected operating points or recorded fixed thresholds.                                  | Post-hoc test-threshold selection is not used for the reported claims.                |
| Feature boundary      | Label source, provenance, and split identifiers are audit fields, not predictive features.                                       | Provenance can explain evidence strength but cannot be used as a shortcut feature.    |
| Metric implementation | All systems report same-work F1, FMR, and HNFMR from the same metric implementation.                                             | Metric comparability does not remove the need for released predictions and checksums. |

## 16 Evaluation Protocol Boundary

The main manuscript summarizes the experimental questions in prose. Table 21 preserves the label stratum, metric, and interpretation boundary for each question so that evidence strength is not mixed across sources.

Table 21: Evaluation protocol. Each question is tied to a label stratum and a metric so that evidence strength is not mixed across sources.

| RQ  | Evidence layer           | Metric        | Interpretation               |
|-----|--------------------------|---------------|------------------------------|
| RQ1 | Gold identity pairs      | Same-work F1  | Duplicate matching ability.  |
| RQ2 | Silver hard negatives    | HNFM          | False-merge safety.          |
| RQ3 | Mechanism analysis       | FMR and HNFM  | Role of risk signals.        |
| RQ4 | Gold/proxy/silver strata | Split metrics | Provenance-aware robustness. |

## 17 Claim-Evidence Matrix

Table 22 maps the major manuscript claims to the evidence required for responsible interpretation. The matrix is part of the submission boundary: a claim should be weakened or removed if the corresponding evidence file is absent from the released artifact package.

Table 22: Claim-evidence matrix for the main manuscript.

| Claim                                                                      | Required evidence                                                                                                                                 | Boundary                                                                                                                                                   |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identity-agenda confusion is a practical false-merge risk.                 | Open-v2 hard-negative evaluation with FMR and HNFM fields.                                                                                        | Supports a targeted scholarly deduplication failure mode, not a prevalence estimate for all domains.                                                       |
| IAD-Risk suppresses hard-negative false merges under the reported setting. | IAD-Risk prediction files, metric summaries, split identifiers, thresholds, and checksums.                                                        | Supports targeted false-merge suppression; does not establish broad method superiority.                                                                    |
| IAD-Bench is provenance-aware.                                             | Pair schema, label strength, label source, provenance fields, and fixture rebuild commands.                                                       | Supports a benchmark contract; does not turn silver or proxy labels into human gold.                                                                       |
| Repository-level reproduction is possible without committing raw data.     | L0/L1 fixture rebuild, public-release scan, unit tests, and documented public-source commands.                                                    | Verifies code and schema behavior; full numerical reproduction requires L2/L3 source files or artifacts.                                                   |
| Cluster-level quality is claimed only after pair-to-cluster audit.         | <code>cluster_metric_summary</code> , <code>cannot_link_audit</code> , cluster assignments, cannot-link coverage, and cluster contamination rate. | Supports cluster-level quality claims only when the released artifact package includes these files with checksums; otherwise the claim remains pair-level. |

Table 23: Claim-evidence boundary used to interpret the reported results. Each claim class is paired with the supporting evidence and the claims that are intentionally not made.

| Claim class                                                            | Required support                                                                                                             | Boundary not claimed                                                                    |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Identity-agenda confusion is a concrete false-merge pathway.           | Hard-negative false-merge rate on agenda-level silver pairs, compared with single-space scientific representation baselines. | A prevalence estimate for all scholarly corpora.                                        |
| IAD-Risk reduces risky automatic merges in the reported setting.       | Held-out Open-v2 IAD-Risk predictions, thresholds, metric summaries, and checksums.                                          | A broad method-ranking claim or a claim that every source distribution is covered.      |
| IAD-Bench supports provenance-aware evaluation.                        | Pair schema fields for relation type, label source, label strength, provenance, split, and hard-negative level.              | A claim that proxy or silver rows are human-gold labels.                                |
| Repository-level reproduction is possible without committing raw data. | Public code, fixture tests, documented data-processing commands, manifest rules, and PDF build scripts.                      | Full numerical reproduction without public source files or a released artifact package. |

Table 24: Result interpretation guardrails for the Open-v2 evidence snapshot. The table separates direct evidence, mechanism-supported reading, and unsupported reading.

| Result row family             | Directly supported reading                                                                                                                                                           | Unsupported reading                                                                                                       |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Representation baselines      | The representation rows test false-merge exposure when one-space scientific relatedness is used as merge evidence.                                                                   | They do not define the best possible representation-threshold system under a fully tuned, same-scope comparative ranking. |
| RoBERTa pair classifier       | The RoBERTa row is a strong supervised comparator that shows pair classification can suppress many hard-negative merges.                                                             | It is not a component-level explanation of identity-agenda risk control.                                                  |
| IAD-Risk transformer variants | The IAD-Risk rows test split-held-out risk gating with explicit identity, agenda, and ANI signals.                                                                                   | The rows are not a claim of broad method superiority without same-scope predictions, intervals, and artifact checksums.   |
| Zero-HNFMR IAD-Risk rows      | Zero observed HNFMR means no observed false merge in the agenda-hard-negative stratum under the reported scope and operating point, while ordinary FMR is still reported separately. | A zero HNFMR value is not evidence that all non-identity false merges are absent or that risk is zero across sources.     |
| Whole table                   | The table gives a mechanism-supported reading that semantic relatedness alone can be unsafe and risk gating can block hard-negative merges in the reported setting.                  | The table is not a same-scope comparative ranking and is not evidence of threshold stability or zero risk.                |

Table 25: Scope compatibility for interpreting the Open-v2 evidence snapshot. The table states what each row family supports and which stronger comparison remains outside the current evidence.

| Row family                    | Pair scope                                                   | Supported interpretation                                                                      | Not supported                                                                                      |
|-------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Representation baselines      | Full available Open-v2 pair scope.                           | Tests whether one-space scientific relatedness creates hard-negative merge risk.              | A tuned neural pair-classifier ranking under the same held-out prediction file.                    |
| RoBERTa pair classifier       | Full available Open-v2 pair scope.                           | Shows that a strong supervised pair classifier can suppress most hard-negative merges.        | A component-level explanation of identity-agenda risk control.                                     |
| IAD-Risk transformer variants | Held-out Open-v2 test scope.                                 | Tests whether explicit identity, agenda, and ANI heads support risk-gated held-out decisions. | A general ranking over every baseline before all rows share the same released prediction scope.    |
| Future stronger comparison    | Same released prediction scope plus manual-validation slice. | Would support a stricter comparative ranking and stronger label-precision claims.             | A claim that this stronger comparison has already been completed by the current evidence snapshot. |

## 18 Uncertainty and Ablation Requirements

Table 26: Threshold and uncertainty reporting protocol. The protocol separates what is required for the current main table from what is required for stronger follow-up claims.

| Item               | Reporting purpose                                                                                                                                             | Interpretation role                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Merge thresholds   | Record $\tau_w$ , $\tau_n$ , $\tau_r$ , validation source, and fixed operating point.                                                                         | Prevents post-hoc threshold selection on the test set.                          |
| Metric uncertainty | Report bootstrap intervals when prediction files and resampling logs are present in the artifact package.                                                     | Distinguishes stable effects from small numerical differences.                  |
| Risk metrics       | Always report FMR and HNFMR with same-work F1.                                                                                                                | Prevents a high F1 score from hiding unsafe hard-negative merges.               |
| Ablation claims    | Claim component causality only when protocol-variant outputs for no-risk-gate, no-ANI-head, single-space, no-cannot-link, and post-hoc-threshold are present. | Separates mechanism-consistent evidence from complete causal ablation evidence. |
| Artifact audit     | Tie each numerical table to predictions, configs, logs, checksums, and commit identifiers.                                                                    | Makes result-level reproduction possible without committing raw data.           |

Table 27: Ablation acceptance protocol. The protocol defines when a component-ablation output is strong enough to support causal interpretation rather than exploratory diagnosis.

| Requirement         | Accepted evidence                                                                                                                                   | Rejected interpretation                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Required variants   | no-risk-gate, no-ANI-head, single-space, no-cannot-link, and post-hoc-threshold outputs are all present in the <code>protocol_variant</code> field. | Component causality from only one removed head or one threshold sweep.                                                                              |
| Scope parity        | Each variant uses the same pair scope, split field, label stratum, and metric implementation as the full IAD-Risk row.                              | Comparing a full-scope baseline with a held-out ablation as if they shared a denominator.                                                           |
| Decision trace      | Each variant releases prediction rows, threshold logs, same-work F1/FMR/HNFMR denominators, and merge/block/defer decisions.                        | Reporting only aggregate metrics without row-level decisions and denominator records.                                                               |
| Artifact binding    | Each variant records configuration, command log, random seed when applicable, code commit, manifest entry, and checksum.                            | Treating unversioned local runs as auditable ablation evidence.                                                                                     |
| Interpretation rule | Accepted non-post-hoc variants may support component-causality statements within the released scope.                                                | Variants with changed pair universe, threshold-selection source, prediction schema, or post-hoc threshold selection remain exploratory diagnostics. |

Table 28: Statistical interpretation boundary for the Open-v2 evidence snapshot. The table states how reported numbers may be read before interval and significance artifacts are released.

| Interpretation item      | Current manuscript status                                                   | Required evidence before stronger claim                                                                                                                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Point estimates          | Reported for the fixed Open-v2 scope, split, and operating points.          | Prediction files, metric summaries, thresholds, and checksums for the same rows.                                                                                                                                                                                    |
| Confidence intervals     | Not reported as primary evidence in the current manuscript.                 | <code>bootstrap_intervals</code> generated from the same prediction files, with command logs and random seeds.                                                                                                                                                      |
| Statistical significance | Not claimed for baseline ranking or model superiority.                      | Predefined tests, multiplicity handling, input artifacts, and reproducible analysis logs.                                                                                                                                                                           |
| Zero HNFMR rows          | Interpreted as no observed hard-negative false merge in the reported scope. | Hard-negative false-merge numerator = 0, HNFMR denominator, hard-negative label stratum, evaluated split, threshold source, and prediction-file checksum before numerical audit; larger source-heldout evidence and manual validation before any zero-risk wording. |
| Model ranking            | Restricted to mechanism interpretation rather than a universal ranking.     | Same-scope predictions, interval estimates, ablations, and manual-validation slice.                                                                                                                                                                                 |

## 19 Decision-to-Metric Mapping

The main manuscript summarizes the decision-to-metric mapping in prose. Table 29 preserves how merge, block, and defer outputs are interpreted by same-work F1, FMR, and HNFMR.

Table 29: Decision-to-metric mapping for selective merge outputs. The mapping prevents block and defer decisions from being hidden inside false-merge metrics.

| Decision output | Metric treatment                                                                                        | Interpretation boundary                                                                                                       |
|-----------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Merge           | Positive prediction for same-work F1; counted as an error for FMR/HNFMR when the label is non-identity. | Measures automatic merge behavior under the fixed operating point.                                                            |
| Block           | Non-merge prediction.                                                                                   | Suppresses false merges but does not count as resolving a positive same-work pair.                                            |
| Defer           | Non-merge prediction.                                                                                   | Protects against unsafe automatic decisions but shifts the case to review and lowers recall when the pair is truly same-work. |

## 20 Metric Formula Boundary

The main manuscript summarizes metric formulas in prose. Table 30 preserves the denominator and label-stratum boundary used to interpret same-work F1, FMR, and HNFMR.

Table 30: Metric formula boundary. Each metric is tied to a denominator and a label stratum so that missing labels, deferrals, and hard-negative rows are not mixed silently.

| Metric       | Formula or denominator                                                                                | Boundary                                                               |
|--------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Same-work F1 | $2TP/(2TP + FP + FN)$ over gold identity rows.                                                        | Defer and block decisions on true same-work rows enter FN.             |
| FMR          | Automatic merges among all non-identity rows divided by all non-identity rows in the evaluated scope. | Measures unsafe automatic merges, not manual-review workload.          |
| HNFMR        | Automatic merges among agenda-level hard negatives divided by the agenda-level hard-negative subset.  | Measures identity-agenda false merges under the hard-negative stratum. |

## 21 Operating Point Disclosure

The main manuscript summarizes operating-point disclosure in prose. Table 31 preserves how each Open-v2 row family obtains its decision field and which artifact is required for threshold audit.

Table 31: Operating point disclosure for the Open-v2 result table. The table records how merge decisions are produced and what artifact is required for threshold audit.

| Row family                      | Decision field                                               | Operating point source                                                                                        | Audit requirement                                       |
|---------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Representation cosine baselines | Cosine score threshold on scientific document embeddings.    | Fixed baseline metric summary; threshold chosen before final test reporting.                                  | Score file, metric summary, and threshold entry.        |
| RoBERTa pair classifier         | Pair match probability threshold.                            | Fixed pair-classifier metric summary under the same pair schema.                                              | Prediction file, metric summary, and model log.         |
| IAD-Risk transformer variants   | $p_{work}$ , $p_{ani}$ , and $p_{risk}$ under the risk gate. | Model configuration; default $\tau_w = \tau_n = \tau_r = 0.5$ unless the artifact config records an override. | Prediction file, model JSON, thresholds, and checksums. |

Default-threshold audit rule. A default IAD-Risk operating point is auditable only when the artifact row records `threshold_source=predeclared_default`, a pre-run configuration checksum, a configuration timestamp or run identifier, and evidence that threshold values were fixed before held-out scoring. The default threshold is a fixed rule, not evidence of validation-selected, optimized, or threshold-stable performance without `threshold_selection_logs` and `threshold_sensitivity_grid` from the same prediction file.

## 22 Threshold Sensitivity Evidence Status

The main manuscript summarizes the threshold-sensitivity evidence status in prose. Table 32 preserves the audit items required before fixed-threshold results can be upgraded to threshold-stability claims.

Table 32: Threshold sensitivity evidence status. The table separates the current fixed-threshold claim from the additional artifact evidence required for threshold-stability claims.

| Audit item              | Current manuscript status                                                                               | Required artifact before stronger claim                                                                  |
|-------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Fixed operating point   | Reported through the operating-point disclosure table and tied to validation or default thresholds.     | Configuration file recording $\tau_w$ , $\tau_n$ , $\tau_r$ , baseline thresholds, and selection source. |
| Threshold grid          | The threshold grid is not reported as primary evidence in the current manuscript.                       | Sensitivity table generated from the same prediction files over predefined threshold ranges.             |
| Metric stability        | Interpreted only for the reported operating points.                                                     | Per-threshold F1, FMR, HNFMR, pair counts, random seeds, and command logs.                               |
| Artifact manifest       | Required for numerical audit but not included in Git.                                                   | Manifest and checksums that bind prediction files, summaries, thresholds, and code commit.               |
| Interpretation boundary | Supports fixed-threshold false-merge control, not threshold-stable ranking across all operating points. | Stronger threshold-stability wording only after the released artifact contains the grid evidence.        |

## 23 Validity Threats Boundary

The main manuscript summarizes validity threats in prose. Table 33 preserves the full validity-threats matrix for audit. The table is intentionally conservative: a mitigation reduces a specific validity risk, but it does not turn proxy or silver evidence into human-adjudicated truth.

Table 33: Threats to validity and claim boundaries. Each risk is paired with the current mitigation and the remaining statement that should not be over-claimed.

| Validity dimension   | Main risk                                                     | Current mitigation                                                                | Remaining boundary                                                                                                             |
|----------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Construct validity   | Silver hard negatives may contain metadata noise.             | Label strength is preserved; silver rows remain stress-test evidence.             | Not a replacement for manual non-identity gold.                                                                                |
| Internal validity    | Threshold choice or split leakage can inflate metrics.        | Operating points, split fields, and pair-leakage audits are required.             | Numerical claims need released predictions and logs.                                                                           |
| External validity    | Open-v2 may not cover every source, topic, or version policy. | Claims are tied to Open-v2 and source-heldout evidence boundaries.                | Broad deployment requires domain-diverse validation.                                                                           |
| Conclusion validity  | HNFMER evidence may be mistaken for full causality.           | Ablation requirements precede component-level causal claims.                      | Complete causality requires no-risk-gate, no-ANI-head, single-space, no-cannot-link, and post-hoc-threshold protocol variants. |
| Reproducibility      | Raw data and large predictions are not stored in Git.         | Public commands, fixtures, manifests, checksums, and artifact rules are provided. | Full numeric audit requires L2/L3 artifacts.                                                                                   |
| Operational validity | Pair errors can propagate through transitive merges.          | Risk gating and cannot-link-aware clustering are described.                       | Cluster guarantees need complete cannot-link coverage.                                                                         |

Result packages that report confidence intervals should include bootstrap outputs generated from the exact prediction files used in the manuscript tables. The repository provides *run-iad-evidence-bootstrap* for stratified pair-level intervals and *run-bootstrap* for general metric intervals. A numerical interval should be cited only when the corresponding CSV file, command log, random seed, and checksum are present in the artifact package.

Threshold-stability claims should be tied to a released threshold-sensitivity grid generated from the same prediction files as the main tables. The grid should report threshold ranges, selected operating points, pair counts, same-work F1, FMR, HNFMER, random seeds, command logs, and checksums. Without this file, the manuscript should discuss only the fixed operating points reported in the main table.

Mechanism claims should be tied to explicit ablation outputs. The repository provides *run-iad-ablation-suite* for comparing full IAD-Risk decisions with protocol variants that remove risk gating, agenda-non-identity blocking, single-space separation, cannot-link blocking, or replace the fixed operating point with a post-hoc threshold diagnostic. The summary files expose the manuscript-facing names in `protocol_variant`; the post-hoc-threshold row is retained as a threshold-overfitting diagnostic and not as standalone causal evidence. Without those outputs, the main manuscript should keep the mechanism claim at the level of stratified HNFMER evidence rather than claiming a complete ablation study.

Cluster-level claims should be tied to artifacts generated after the pair decisions are converted

into connected components. A defensible package should include cluster assignments, a cluster metric summary, a cannot-link audit, and command logs for the merge policy. Without those outputs, pair-level FMR and HNFMR should not be interpreted as proof that cluster-level contamination has been reduced or eliminated.

## 24 Claim Interpretation Boundary

The main manuscript summarizes the claim interpretation boundary in prose. Table 34 preserves the full matrix that maps each review dimension to the present evidence and the wording boundary before stronger claims.

Table 34: Claim interpretation boundary. Each row maps a common review dimension to the present evidence and the wording boundary that should not be crossed without additional artifacts.

| Review dimension        | Main evidence location                                                              | Supported wording                                                                                                 | Boundary before stronger wording                                                                                            |
|-------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Contribution clarity    | Introduction, problem formulation, and IAD-Bench contract.                          | The manuscript identifies identity-agenda confusion and HNFMR as a false-merge safety problem.                    | Do not frame the work as a universal replacement for entity resolution or scientific representation learning.               |
| Writing reproducibility | Method, implementation details, data/code availability, and supplementary protocol. | The repository supports code-level checks, fixture rebuilds, schema validation, and artifact-release preparation. | Full numerical audit requires external artifacts with predictions, logs, manifests, and checksums.                          |
| Experimental strength   | Open-v2 evidence snapshot, operating-point disclosure, and result audit trail.      | Current evidence supports targeted false-merge suppression under the stated Open-v2 scope.                        | Stronger ranking claims require same-scope predictions, threshold logs, bootstrap intervals, and released metric tables.    |
| Evaluation completeness | Claim-evidence boundary, manual-validation boundary, and uncertainty requirements.  | The manuscript separates gold, proxy, silver, and future manual-review evidence layers.                           | Component causality or human-gold claims require artifact-backed ablations, threshold grids, and a manual-validation slice. |
| Method design soundness | Risk score rationale, merge-decision rule, and validity threats.                    | IAD-Risk exposes identity, agenda, ANI risk, thresholds, and cannot-link behavior.                                | Broad deployment claims require source-heldout validation, topic-heldout checks, and failure-case analysis.                 |

## 25 Reviewer Evidence Gate

This section states the evidence gate that should be applied before a reviewer reads the present results as stronger than the manuscript claims. The current package supports a bounded mechanism claim: identity-agenda separation and false-merge risk gating can suppress hard-negative automatic merges in the reported Open-v2 evidence snapshot. Stronger claims require released artifacts that bind the same rows, thresholds, logs, and checksums to the submitted text.

Table 35: Reviewer evidence gate for upgrading manuscript claims. Each stronger interpretation requires an explicit artifact source before it should appear in the abstract, conclusion, cover letter, or response to reviewers.

| Review dimension        | Current supported reading                                                                              | Evidence required before stronger wording                                                                                           |
|-------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Contribution clarity    | Identity-agenda confusion is a targeted scholarly false-merge failure mode.                            | Broader task-replacement claims require source diversity, same-scope prediction files, and domain-facing error analysis.            |
| Experimental strength   | Open-v2 supports fixed-scope false-merge control under the reported operating points.                  | Model-ranking claims require same-scope prediction files, threshold logs, bootstrap intervals, and released metric tables.          |
| Evaluation completeness | The manuscript separates gold, proxy, silver, and future manual evidence layers.                       | Component-causality or label-precision claims require an ablation suite, threshold grid, and manual-validation slice.               |
| Method design soundness | Risk gating exposes identity, agenda, ANI, threshold, and cannot-link behavior.                        | Deployment claims require source-heldout validation, topic-heldout checks, review-load analysis, and cluster-level artifact audits. |
| Reproducibility         | Git supports code-level checks, fixture rebuilds, schema validation, and artifact release preparation. | Full numerical audit requires an external release with manifests, checksums, command logs, and fixed commit identifiers.            |

The practical rule is a do not upgrade claims rule: do not strengthen the title, abstract, conclusion, cover letter, or reviewer response beyond the bounded Open-v2 mechanism claim unless the corresponding evidence row in Table 35 is satisfied by released artifacts.

## 26 Mechanism Evidence Boundary

The main manuscript summarizes the mechanism-evidence boundary in prose. Table 36 preserves the mechanism questions, current evidence, and interpretation boundary needed before stronger causal or deployment claims are made.

Table 36: Mechanism evidence and interpretation boundary. The table separates current evidence from ablation outputs that would be required for a stronger causal claim.

| Mechanism question                                       | Current evidence                                                                        | Interpretation boundary                                                                                                                                |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does topical relatedness create merge risk?              | Representation baselines show high false-merge rates on OpenAlex silver hard negatives. | Supports a targeted failure mode, not a universal prevalence estimate.                                                                                 |
| Does explicit risk gating suppress hard-negative merges? | IAD-Risk transformer variants keep low HNFMR under the reported Open-v2 split.          | Supports the reported contract and thresholds; transfer must be rechecked on new source mixes.                                                         |
| Is the gain caused by each IAD-Risk component?           | The current manuscript reports mechanism-consistent HNFMR evidence.                     | A full causal claim requires protocol-variant outputs for no-risk-gate, no-ANI-head, single-space, no-cannot-link, and post-hoc-threshold diagnostics. |
| Can cluster-level contamination be eliminated?           | The method supports cannot-link-aware constrained union-find.                           | Guarantees require sufficient cannot-link coverage and cluster-level artifact audits.                                                                  |

## 27 Error Taxonomy Boundary

The main manuscript summarizes the identity-agenda error taxonomy in prose. Table 37 preserves the full taxonomy, the IAD-Risk control associated with each failure mode, and the audit evidence needed before stronger error-distribution claims are made.

Table 37: Error taxonomy for identity-agenda confusion. The table identifies the expected failure mode, the IAD-Risk control, and the audit evidence needed for stronger claims.

| Error category                    | Typical source of confusion                                                                  | IAD-Risk control                                                                     | Stronger audit evidence                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Same task, different contribution | Papers share a benchmark, model family, or problem statement.                                | Agenda-non-identity head and HNFMR reporting.                                        | Human-reviewed hard-negative slice with pair notes.              |
| Citation-neighborhood neighbor    | Papers share references or citation community but are distinct works.                        | Reference-overlap features contribute to agenda evidence, not direct merge evidence. | Citation-derived pair provenance and checksum-fixed predictions. |
| Version or extension boundary     | Preprint, short paper, conference, and journal records are similar but may not be mergeable. | Conservative defer/block behavior when identity evidence is incomplete.              | Manual adjudication with version-policy labels.                  |
| Identifier conflict               | DOI, arXiv, or OpenAlex identifiers disagree despite text similarity.                        | Different-identifier conflicts enter the ANI risk head and cannot-link logic.        | Identifier-resolution logs and conflict audit table.             |
| Sparse metadata                   | Abstract, venue, author, or identifier fields are missing or noisy.                          | Merge requires positive identity evidence rather than semantic similarity alone.     | Missing-field stratified metrics and per-source diagnostics.     |

## 28 Manual Validation Protocol

Manual validation is a future evidence layer for assessing label precision and difficult boundary cases; it is not completed in the current manuscript package. A defensible validation slice should sample 500–1,000 pairs across the strata most likely to affect claim strength: OpenAlex/OpenCitations silver hard negatives, high-score representation false-merge candidates, IAD-Risk blocked or deferred pairs, version-boundary cases, identifier conflicts, and sparse-metadata cases.

Table 38: Manual validation boundary for interpreting silver hard negatives. The current manuscript reports stress-test evidence and reserves human-validation claims for a released reviewed slice.

| Evidence item          | Current role                                                                | Requirement before stronger claim                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Silver hard negatives  | Stress-test evidence for agenda-related false-merge risk.                   | A 500–1,000 pair reviewed slice covering hard negatives, high-score false-merge candidates, blocked or deferred pairs, version boundaries, identifier conflicts, and sparse metadata. |
| Human review process   | Specified as a future validation layer rather than current result evidence. | The protocol requires two independent reviewers to inspect records while blind to model scores and merge decisions.                                                                   |
| Adjudication artifacts | Not present in the current Git package or main result table.                | The release must include an adjudication log, an agreement report, pair-level notes, and checksums before claiming label precision.                                                   |
| Manuscript claim       | Supports bounded false-merge stress testing under Open-v2.                  | The paper does not claim complete human validation or that silver rows replace human-gold non-identity labels.                                                                        |

Table 39: Manual validation protocol required before stronger label-precision claims.

| Protocol item  | Requirement                                                                                                                                              | Claim boundary                                                                                           |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Sampling       | Stratified 500–1,000 pair slice across silver hard negatives, model-disagreement pairs, version-boundary pairs, and sparse-metadata pairs.               | The slice estimates difficult-case precision; it is not a prevalence estimate for all scholarly records. |
| Review process | two independent reviewers inspect titles, authors, venues, identifiers, abstracts, and citation context while blind to model scores and merge decisions. | Model outputs cannot determine the human label.                                                          |
| Label schema   | Reviewers choose same work, agenda non-identity, unrelated, version boundary, or uncertain, and record a short evidence note.                            | Uncertain cases remain separate and are not forced into gold labels.                                     |
| Adjudication   | Disagreements are resolved by an adjudication log that records the final relation, rationale, and reviewer IDs or anonymous reviewer codes.              | Adjudicated labels become auditable only when the log is released.                                       |
| Reporting      | The artifact package reports inter-annotator agreement, per-stratum counts, conflict rates, and checksums for the reviewed slice.                        | The manuscript must not claim human gold or stronger label precision until these files exist.            |

## 29 Data and Code Availability Boundary

The main manuscript states the data and code availability boundary in prose. Table 40 preserves the full availability matrix that separates version-controlled source files from public-source inputs and released result artifacts.

Table 40: Data and code availability boundary. The table separates what is version-controlled in Git from what must be obtained from public sources or a released artifact package.

| Availability class                          | Included location                                                       | Evaluation use                                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source code and CLI entry points            | Project repository.                                                     | Supports installation, schema checks, fixture rebuilds, baseline execution, model training entry points, and manuscript builds.                                                                              |
| Small public fixtures and schema contracts  | Project repository.                                                     | Supports L0/L1 code-level reproduction without downloading full raw datasets.                                                                                                                                |
| Raw third-party source files                | Excluded from Git; obtained from original providers.                    | Raw third-party data remain governed by original provider licenses and access conditions; released artifacts should not redistribute raw provider files unless source terms explicitly allow redistribution. |
| Full prediction files and model checkpoints | Excluded from Git unless provided through an external artifact release. | Prevents large local outputs from being treated as repository source files.                                                                                                                                  |
| Derived evaluation artifacts                | External artifact release.                                              | Supports L2/L3 result-level audit through source input manifests, processing run logs, prediction files, threshold logs, manifests, checksums, and commit identifiers.                                       |

Full numerical reproduction requires public-source rebuilds or released artifacts; the Git repository alone supports code-level checks, fixture behavior, schema validation, and manuscript builds. The release should expose derived outputs, source input manifests, processing run logs, and checksums rather than raw provider files unless the original source terms explicitly permit redistribution.

## 30 Claim Boundary

The main manuscript treats DeepMatcher-style labels as gold identity evidence. It treats SciRepEval and SciDocs-style proximity as same-agenda proxy evidence. It treats OpenAlex and OpenCitations-derived pairs as silver hard-negative evidence. OpenAlex and OpenCitations signals are not human gold labels. Cluster-level quality is not claimed unless the released artifact package includes cluster assignments, `cluster_metric_summary`, `cannot_link_audit`, `cannot_link` coverage, and cluster contamination rate. Broad method-ranking claims require stronger source-heldout evidence and manual validation beyond the current manuscript package.